Circuit Debugging Assistant
Circuit Diagnostic Report

Report ID: ANALYSIS-0324A20A
Date: 2026-10-13 23:48
Circuit: Rc Low Pass

STATUS:
POTENTIAL ISSUES IDENTIFIED

1. Uploaded versus Expected Circuit

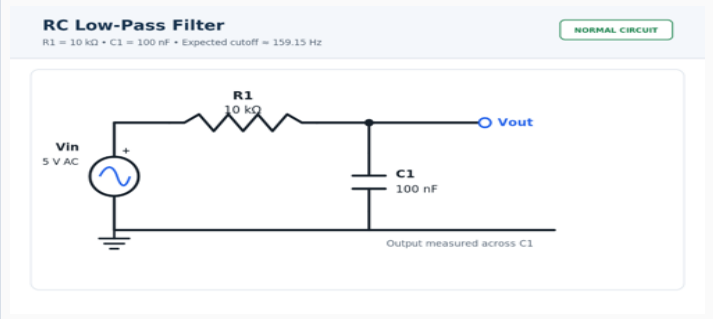


Figure 1A: Uploaded Circuit Schematic

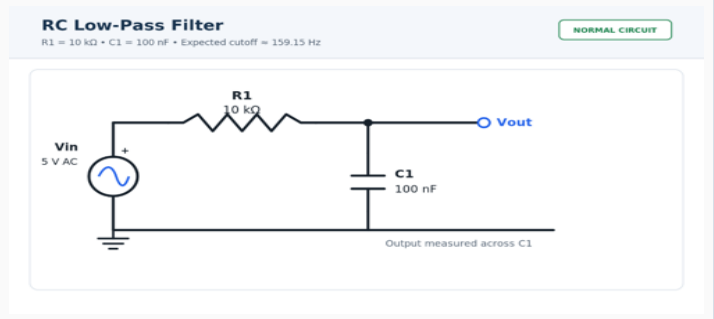


Figure 1B: Expected / Reference Circuit Model

2. Diagnosis Summary

The device under test was analyzed as a **Rc Low Pass**. Numerical multi-evidence evaluation identified a probable parameter shift on **R -20%** (normalized residual score: **29.9877**). A total of 0 structural or parameter observations were cataloged.

3. Detected Flaws & Consistency Register

Severity	Component / Net	Finding & Evidence	Recommended Action
Major	**R**	Normalized residual S(f)=29.988 matches R -20% signature.	Verify R value with bench meter or replace with nominal part.

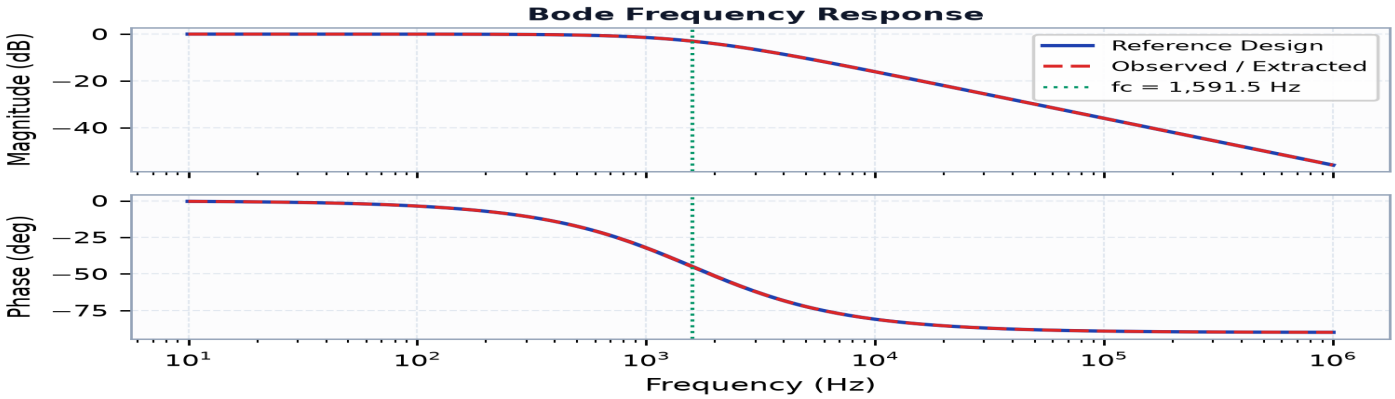
4. Confirmed Component Values

Reference	Component Type	Extracted / Nominal Value	Tolerance	Verification Status
VIN	Input Voltage Source	1 V	±1.0%	Confirmed
R	Film Resistor	1000 Ω	±5.0%	Confirmed
C	Ceramic / Electrolytic Capacitor	100.00 nF	±10.0%	Confirmed

5. Expected Behaviour & Governing Calculations

- **Transfer Function:** H(f) = 1 / (1 + j 2π f R C)
- **Theoretical Cutoff Frequency:** fc = 1 / (2π · R · C) = **1591.55 Hz**
- **Initial Step Current (Independent Observable):** I_step(0+) = Vin / R = **1.00 mA**

6. Numerical Simulation Curves



7. Recommended Engineering Actions

1. **Physical Verification:** Perform **DMM Resistance Measurement on R1** to independently separate ambiguous parameter hypotheses.
2. **Signal Integrity:** Measure test points with a 10:1 passive probe to avoid probe capacitance loading.
3. **Post-Repair Gate:** Confirm that measured response stays within ±5% of reference cutoff before releasing design.